

# Ali Alemami

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**Computer Science student specializing in backend & low-level systems engineering**, with hands-on experience in **C/C++** (POSIX system APIs, multithreading, concurrency, memory models) and **C#.NET** (3-tier architecture, ADO.NET, SQL Server). Seeking an entry-level backend, systems, or software engineering role.

## EDUCATION

42 Amman — Common Core, 22/24 projects complete (*peer-reviewed, project-based*)

2025 – present

The Hashemite University — BSc in Computer Science

2022 – 2026 (Expected)

## TECHNICAL SKILLS

|                                    |                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>      | C (POSIX/C99), C++ (98/11/17), C# (.NET), SQL (T-SQL), Bash                                                                  |
| <b>Systems &amp; OS Concepts:</b>  | POSIX APIs, Multithreading (pthreads), Concurrency & Mutexes, Sockets / epoll, Process Control & IPC, Memory (Valgrind/ASan) |
| <b>CS Fundamentals:</b>            | Data Structures & Algorithms, OOP & Design Patterns, 3-Tier Architecture, Computer Networks                                  |
| <b>Tools &amp; Infrastructure:</b> | SQL Server, ADO.NET, Docker, Docker Compose, NGINX, Git, Linux, GDB, Makefiles                                               |
| <b>Languages (Spoken):</b>         | Arabic (Native), English (Fluent / Professional)                                                                             |

## 42 CORE CURRICULUM — 22/24 PROJECTS COMPLETE

- **Multithreading & Concurrency (C — philo)**: Engineered a race-free simulation of Dijkstra's Dining Philosophers using POSIX threads (pthreads) and mutex locks; built dedicated asynchronous monitoring routines and microsecond-precision delta timers (gettimeofday) preventing thread starvation and deadlocks.
- **HTTP Web Server (C++98 — webserv, in progress)**: Engineering a high-performance non-blocking HTTP/1.1 server using epoll I/O event multiplexing; implementing stateful request stream parsing (GET/POST/DELETE), NGINX-style configuration parsing, client timeout handling, and CGI execution pipelines.
- **DevOps & Container Infrastructure (inception)**: Architected a multi-container infrastructure with Docker Compose on Alpine Linux; configured isolated containers for NGINX with TLS/SSL encryption (v1.2/v1.3), WordPress (PHP-FPM), and MariaDB with dedicated bind mounts, internal bridge networking, and PID 1 entrypoint signal management.
- **Unix Shell & AST Engine (C — minishell)**: Implemented a POSIX-compliant shell with an Abstract Syntax Tree (AST) parser to enforce operator precedence ((), &&, ||), nested subshell forks, quote-aware tokenization, heredocs with delimiter expansion, variable expansion, and signal handling; zero fd/memory leaks via Valgrind.
- **3D Graphics & Raytracing Engine (C — mini\_rt)**: Built a 3D raytracing renderer from scratch in C; implemented vector math (dot/cross products, normalization), quadratic discriminant ray-surface intersections (spheres, planes, cylinders), and Phong reflection models (ambient, diffuse, specular highlights, and hard shadows) rendered via MiniLibX.
- **C++ Modules & Ford-Johnson Algorithm (C++ — cpps)**: Completed 42 C++ Modules 00–09 (OOP, Orthodox Canonical Form, polymorphism, templates, STL); implemented the **Ford-Johnson Merge-Insert Sort (CPP09)** optimizing comparison steps using dual STL sequence containers (std::vector, std::deque) and Jacobsthal sequence pairing heuristics.
- **Foundations & Algorithms: push\_swap** (bitwise radix sort across two stacks), **pipex** (multi-pipe fork/dup2), **fract-ol** (Mandelbrot/Julia sets), **minitalk** (UNIX signal bit-shift IPC), **libft**, **ft\_printf**, **get\_next\_line**. *In progress: transcendence* (real-time multiplayer web app).

## PROGRAMMINGADVICES ROADMAP — 17/24 COURSES COMPLETE

*Dr. Mohammed Abu-Hadhoud curriculum — software architecture, advanced problem solving, database engineering, and .NET*

- **Bank & ATM CLI System (C++ — Bank-and-ATM-System-CPP)**: Built a terminal-based banking and ATM system in C++ with bitmask user permissions (bitwise flags), transaction audit logging, and file-based data persistence.
- **Contacts Management App (C#, ADO.NET, SQL Server — Contacts-App-3Tier)**: Developed a 3-tier desktop application separating UI, business logic, and data access layers; utilized parameterized T-SQL queries to ensure data integrity and prevent SQL injection.
- **Relational Database Schemas & SQL (SQL Server — Programming-Advices-Core)**: Designed 5 relational database schemas (Medical Clinic Booking, E-Commerce Store, HR/Payroll, Vehicle Registry, Employee Analytics) with primary/foreign key constraints, normalization, and multi-table analytical JOIN queries.
- **Algorithms & Problem Solving (C++ — 225+ Problems)**: Implemented solutions across 5 progressive difficulty levels for data structures, matrix operations, string parsing, combinatorial problems, and date/time arithmetic algorithms.
- **Windows Forms Projects (C#, .NET — CSharp-WinForms-Collection)**: Built 10+ desktop utility and game applications in C# focusing on UI event handling, input validation, and object-oriented design principles.